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Managing decref processing from tablets containing sorted buckets

Abstract

A technique for managing reference-count decrements (decrefs) includes defining an aggregation domain that includes multiple tablets created at respective points in time, each of the tablets containing multiple sorted buckets arranged in a respective sequence of bucket identifiers that is repeated across all of the tablets, the sorted buckets containing decrefs for respective sets of metadata pages. The technique further includes assigning a decref job to a specified set of bucket identifiers and destaging, by the decref job, multiple decrefs from the tablets to persistent metadata pages, beginning with decrefs in buckets having the specified set of bucket identifiers and proceeding until a target number of decrefs has been destaged.

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Background/Summary

BACKGROUND

(1) Data storage systems are arrangements of hardware and software in which storage processors are coupled to arrays of non-volatile storage devices, such as magnetic disk drives, electronic flash drives, and/or optical drives. The storage processors, also referred to herein as “nodes,” service storage requests arriving from host machines (“hosts”), which specify blocks, files, and/or other data elements to be written, read, created, deleted, and so forth. Software running on the nodes manages incoming storage requests and performs various data processing tasks to organize and secure the data elements on the non-volatile storage devices.

(2) Data storage systems commonly use block sharing for supporting deduplication, snapshots, and the like. In a typical arrangement, logical blocks of various files or volumes point to virtual blocks, which in turn point to physical blocks where data are stored. Because of block sharing, multiple logical blocks may point to the same virtual block. Each virtual block may have a reference count, which counts the number of logical blocks that point to it. A reference count for a virtual block may be incremented in response to copies, new snapshots, and the like, and it may be decremented in response to deletes, for example. Reference count decrements may be referred to herein as

“decrefs.” Once the reference count for a virtual block reaches zero, the associated physical block is no longer being used and the space occupied by the physical block can be reclaimed.

(3) Commercial storage systems produce huge numbers of decrefs in the course of their normal activities. To implement these decrefs, a storage system may process decrefs in batches. In one arrangement, the storage system adds newly-created decrefs to pages in a decref bin. When resources in the storage system become available, the storage system fetches a number of pages from the decref bin and begins processing them, e.g., by modifying virtual blocks to reflect updated reference counts as indicated by the decrefs in the fetched pages.

SUMMARY

- (4) Certain embodiments are directed to a method of managing reference-count decrements (decrefs). The method includes defining an aggregation domain that includes multiple tablets created at respective points in time, each of the tablets containing multiple sorted buckets arranged in a respective sequence of bucket identifiers that is repeated across all of the tablets, the sorted buckets containing decrefs for respective sets of metadata pages. The method further includes assigning a decref job to a specified set of bucket identifiers and destaging, by the decref job, multiple decrefs from the tablets to persistent metadata pages, beginning with decrefs in buckets having the specified set of bucket identifiers and proceeding until a target number of decrefs has been destaged.
- (5) In some examples, destaging multiple decrefs from the tablets includes merging decrefs from buckets having a particular bucket identifier across at least some of the tablets and updating one or more metadata pages based on the merged decrefs.
- (6) In some examples, the decref job is a first decref job, and merging decrefs from the buckets having the particular bucket identifier includes ending the first decref job after merging fewer than all buckets having the particular bucket identifier, in response to the target number of decrefs being reached, and starting a second decref job configured to merge remaining buckets having the particular bucket identifier that were not merged by the first decref job.
- (7) In some examples, destaging, by the decref job, multiple decrefs from the tablets includes completely destaging decrefs from all buckets for each bucket identifier processed by the decref job.
- (8) In some examples, completely destaging all decrefs for each bucket identifier processed by the decref job causes the target number of decrefs to be exceeded, and the method further includes providing feedback to a system scheduler, the feedback indicating a number of decrefs actually destaged by the decref job and enabling the scheduler to compensate when scheduling future jobs.
- (9) In some examples, the set of bucket identifiers includes a plurality of bucket identifiers, and destaging, by the decref job, multiple decrefs includes operating multiple destagers in parallel.
- (10) In some examples, the method further includes tracking an average aggregation of decrefs for each tablet and adjusting the number of tablets in the aggregation domain to achieve a desired level of aggregation.
- (11) In some examples, the aggregation domain includes a number of tablets that is determined based at least in part on an average aggregation of decrefs achieved during one or more previous destaging operations of previously created tablets.
- (12) In some examples, the number of tablets in the aggregation domain is further based at least in part on an amount of free space in a decref storage tier configured to store tablets awaiting destaging.
- (13) In some examples, the method is operated in a data storage system, and the number of tablets in the aggregation domain is further based at least in part on an amount of free space in the data storage system.
- (14) In some examples, the method is operated in a data storage system, and the method further includes establishing a desired number of tablets in the aggregation domain and, before the desired number of tablets has been created, triggering an early destage from the tablets already created,

based on a condition of the data storage system.

(15) In some examples, the condition of the data storage system includes host I/O requests received by the data storage system being idle.

(16) In some examples, the early-destage is triggered based at least in part on an amount of free storage space in the data storage system falling below a low-water mark.

(17) In some examples, the data storage system runs a set of other background jobs that have dependencies on the decref job, and the early-destage is triggered based at least in part on an amount of processing debt from the set of other background jobs exceeding a high-water mark.

(18) Other embodiments are directed to a computerized apparatus constructed and arranged to perform a method of managing reference-count decrements (decrefs), such as the method described above. Still other embodiments are directed to a computer program product. The computer program product stores instructions which, when executed on control circuitry of a computerized apparatus, cause the computerized apparatus to perform a method of managing reference-count decrements (decrefs), such as the method described above.

(19) The foregoing summary is presented for illustrative purposes to assist the reader in readily grasping example features presented herein; however, this summary is not intended to set forth required elements or to limit embodiments hereof in any way. One should appreciate that the above-described features can be combined in any manner that makes technological sense, and that all such combinations are intended to be disclosed herein, regardless of whether such combinations are identified explicitly or not.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) The foregoing and other features and advantages will be apparent from the following description of particular embodiments, as illustrated in the accompanying drawings, in which like reference characters refer to the same or similar parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles of various embodiments.

(2) FIG. 1 is a block diagram of an example environment in which embodiments of the improved technique can be practiced.

(3) FIG. 2 is a block diagram showing an example arrangement in which a decref job is completed before completely processing a column of buckets.

(4) FIG. 3 is a block diagram showing an example arrangement in which a decref job is completed upon completely processing a column of buckets, but after exceeding a target number of decrefs.

(5) FIG. 4 is a block diagram showing an example arrangement in which a decref job is completed by multiple destagers operating on respective columns of buckets.

(6) FIG. 5 is a block diagram showing example factors that contribute to a determination of an aggregation-domain size.

(7) FIG. 6 is a block diagram showing example factors that contribute to an early destaging of decrefs prior to reaching a desired aggregation-domain size.

(8) FIG. 7 is a flowchart showing an example method of managing decrefs.

DETAILED DESCRIPTION

(9) Embodiments of the improved technique will now be described. One should appreciate that such embodiments are provided by way of example to illustrate certain features and principles but are not intended to be limiting.

(10) Continuing developments in storage-system technology have replaced the decref bin with persistent tablets. The tablets store decrefs in sorted buckets arranged based on virtual blocks. A storage system creates the tablets from in-memory buckets, which receive decrefs as the storage

system produces them. Different tablets capture decrefs collected by the in-memory buckets during respective intervals of time. Altogether, the tablets capture decrefs accumulated over an extended period of time and allow decrefs for virtual blocks to be aggregated before writing them to disk. When the virtual blocks are stored in flash drives, such aggregation extends the lifetime of the flash drives by reducing the number of writes. The ability to aggregate decrefs was sharply limited when using the decref bin, though, as decrefs were not sorted and considerable resources were required to achieve just a small amount of aggregation.

(11) The use of tablets creates new challenges, however. In the prior scheme, each page fetched from the decref bin contained an approximately equal number of decrefs, meaning that background jobs that processed decrefs could complete within predictable amounts of time. Predictable workloads in these jobs can be valuable as they reduce the need to accommodate sudden increases in demand, which can require overprovisioning of computing resources, as well as sudden decreases in demand, which can result in underutilization of provisioned computing resources.

(12) But the number of decrefs stored in tablets is not predictable. Some buckets within the tablets can be full, whereas others can be empty, or nearly empty. Also, the use of tablets introduces the additional factor of aggregation. What is needed, therefore, is a way of leveling the workload of background jobs that process decrefs from tablets, without interfering with the ability to aggregate over arbitrarily large numbers of tablets.

(13) The above need is addressed at least in part with an improved technique of managing reference-count decrements (decrefs). The technique includes arranging decref jobs to process a constant, or nearly constant, number of decrefs per job. A decref job begins with a specified set of bucket identifiers common to multiple tablets across an aggregation domain and processes decrefs from the specified bucket identifiers until a target number of decrefs has been reached, continuing to additional bucket identifiers as needed.

(14) Advantageously, the improved technique provides a consistent workload for decref jobs, even though the numbers of decrefs in different buckets can vary substantially. Also, the improved technique can typically level the workload of decref jobs without impacting aggregation, which is allowed to vary independently of the workload per decref job.

(15) FIG. 1 shows an example environment **100** in which embodiments of the improved technique can be practiced. Here, multiple hosts **110** are configured to access a data storage system **116** over a network **114**. The data storage system **116** includes one or more nodes **120** (e.g., node **120a** and node **120b**), and storage **190**, such as magnetic disk drives, electronic flash drives, and/or the like. Nodes **120** may be provided as circuit board assemblies or blades, which plug into a chassis (not shown) that encloses and cools the nodes. The chassis has a backplane or midplane for interconnecting the nodes **120**, and additional connections may be made among nodes **120** using cables. In some examples, the nodes **120** are part of a storage cluster, such as one which contains any number of storage appliances, where each appliance includes a pair of nodes **120** connected to shared storage. In some arrangements, a host application runs directly on the nodes **120**, such that separate host machines **110** need not be present. No particular hardware configuration is required, however, as any number of nodes **120** may be provided, including a single node, in any arrangement, and the node or nodes **120** can be any type or types of computing device capable of running software and processing host I/O's.

(16) The network **114** may be any type of network or combination of networks, such as a storage area network (SAN), a local area network (LAN), a wide area network (WAN), the Internet, and/or some other type of network or combination of networks, for example. In cases where hosts **110** are provided, such hosts **110** may connect to the node **120** using various technologies, such as Fibre Channel, iSCSI (Internet small computer system interface), NVMeOF (Nonvolatile Memory Express (NVMe) over Fabrics), NFS (network file system), and CIFS (common Internet file system), for example. As is known, Fibre Channel, iSCSI, and NVMeOF are block-based protocols, whereas NFS and CIFS are file-based protocols. The node **120** is configured to receive

I/O requests **112** according to block-based and/or file-based protocols and to respond to such I/O requests **112** by reading or writing the storage **190**.

(17) In an example, the storage **190** includes multiple storage regions. These include a data region **192** for storing user data, a metadata region **194** for storing system metadata, and a dedicated decref tier **196**, for storing tablets **150**, described below. The metadata stored in the metadata region **194** may include virtual blocks **194a**, which contain reference counts **194b**.

(18) The depiction of node **120a** is intended to be representative of all nodes **120**. As shown, node **120a** includes one or more communication interfaces **122**, a set of processors **124**, and memory **130**. The communication interfaces **122** include, for example, SCSI target adapters and/or network interface adapters for converting electronic and/or optical signals received over the network **114** to electronic form for use by the node **120a**. The set of processors **124** includes one or more processing chips and/or assemblies, such as numerous multi-core CPUs (central processing units). The memory **130** includes both volatile memory, e.g., RAM (Random Access Memory), and non-volatile memory, such as one or more ROMs (Read-Only Memories), disk drives, solid state drives, and the like. The set of processors **124** and the memory **130** together form control circuitry, which is constructed and arranged to carry out various methods and functions as described herein. Also, the memory **130** includes a variety of software constructs realized in the form of executable instructions. When the executable instructions are run by the set of processors **124**, the set of processors **124** is made to carry out the operations of the software constructs. Although certain software constructs are specifically shown and described, it is understood that the memory **130** typically includes many other software components, which are not shown, such as an operating system, various applications, processes, and daemons.

(19) As further shown in FIG. 1, the memory **130** “includes,” i.e., realizes by execution of software instructions, an array of in-memory buckets **140** and multiple persistent tablets **150** (e.g., **150-1** through **150-M**), which form an aggregation domain **160** and may be backed by the decref tier **196**. The memory **130** further includes a scheduler **170**, a decref job **180**, and other background (BG) jobs **180a**.

(20) The array of in-memory buckets **140** includes multiple buckets **142** having respective bucket identifiers B1 through BN, where N is a positive integer. Each bucket **142** is arranged to receive decrefs (D) **154** for a respective set of virtual-block pages **194a**. For example, each virtual-block page **194a** has a logical identifier (LI), such as an LBA (Logical Block Address), and virtual-block pages **194a** are assigned to buckets **142** based on LI (e.g., based on hash of LI). Typically, each bucket **142** contains decrefs **154** for multiple LIs, which may be arranged in a tree structure organized by LI. Each time the storage system produces a decref **154** pursuant to some activity, such as a delete, the storage system identifies the LI of the affected virtual-block page **194a**, hashes the LI, and identifies the bucket **142** dedicated to the hash range that includes the hashed LI. The system then stores the decref **154** in the identified bucket.

(21) In some examples, which are not intended to be limiting, the array of in-memory buckets **140** includes two separate arrays (not shown) of in-memory buckets, an active array that receives new decrefs **154** and a frozen array that was previously active but is now frozen. Once the active array becomes full, or in some cases sooner, the storage system freezes the active array so that it becomes the frozen array and clears the frozen array so that it becomes active. The newly-frozen array is then converted to a tablet, e.g., by copying the decrefs from the in-memory buckets **142** to corresponding persistent buckets **152** in a persistent tablet **150**. A new persistent tablet **150** may be created upon each conversion operation, which may correspond to each active-frozen switch. Tablets **150** thus contain decrefs **154** accumulated by the in-memory buckets **140** during respective intervals of time, with each tablet **150** reflecting the decrefs at the point in time when an active array of tablets became frozen.

(22) Each of the tablets **150** contains an array of buckets **152** having bucket identifiers B1 through BN, which correspond, one-to-one, to the bucket identifiers of the in-memory buckets **140**. The

tablets **150** thus reflect the same sorting that was used when first placing decrefs **154** into the in-memory buckets **140**. Corresponding buckets having the same bucket identifier may be referred to herein as a “column” **154** of buckets. Because of the way that decrefs are assigned to in-memory buckets **140**, aggregation of decrefs for virtual-block pages **194a** is possible only within columns **154** but not between columns **154**.

(23) In example operation, the storage system **116** accumulates decrefs **154** in the in-memory buckets **140** and converts the in-memory buckets **140** to persistent tablets **150** at respective points in time (T1, T2, . . . , TN). The storage system may define the aggregation domain **160** as the desired number of tablets **150** over which the storage system may aggregate decrefs, in this case, M tablets. Once the M tablets have been created, the scheduler **170** starts the decref job **180**, which is configured to destage **166** a target number **182** of decrefs from the tablets **150-1** through **150-M**.

(24) Preferably, the decref job **180** destages (**166**) decrefs column **154** by column **154**. For example, the decref job **180** may start with the B1 column, counting all the decrefs for each LI in buckets B1 down all tablets **150-1** through **150-M**. Once the decref job **180** reaches the end of a column (e.g., bucket B1 in tablet **150-1**), the decref job **180** may destage (**166**) the accumulated decrefs **154** from that column to storage **190**. For example, the decref job may create a merged bucket **162**, which includes the total number of decrefs for each LI tracked in the B1 column. The decref job **180** may then perform a read-modify-write operation **164**, which includes reading from storage **190** a virtual-block page **194a** for each LI in the merged bucket **182**, updating the reference count **194b** for each virtual-block page **194a** based on the aggregated decrefs for that virtual-block page, and writing the virtual-block page **194a** back to storage **190**.

(25) If the total number of decrefs counted in the B1 column has reached the target number **182**, then the decref job **180** may stop, having achieved its aim. Typically, though, the decref job **180** will not reach the target number **182** until it has processed multiple columns **154**. In such cases, the decref job **180** may go on to the next column, e.g., the B2 column, and begin processing decrefs from there, counting decrefs as it goes. Once the decref job **180** reaches the end of the B2 column, the decref job **180** performs another destage operation **166**, by creating another merged bucket **162** and performing another read-modify write operation **164**, proceeding this way column-by-column until the target number **182** has been reached.

(26) Once the decref job **180** has completed, the scheduler **170** may start another decref job, fitting it in along with other background jobs **180a**. Each additional decref job may process the same target number **182** of decrefs. In some examples, the target number **182** may be adjusted as circumstances change, but the fact that the target number **182** is consistent ensures that decref processing occurs smoothly, helping the storage system to avoid sudden spikes in demand for resources as well as sudden dips, and improving efficiency.

(27) FIG. 2 shows an example in which the decref job **180** reaches the target number **182** of decrefs before reaching the end of a column **154**. As shown, a first decref job **180a** begins processing decrefs in the B1 column, completing the column and performing a destage operation **166**. But since the target number **182** has not yet been reached, the first decref job **180a** proceeds to the B2 column, and then to the B3 column. Upon counting the decrefs in bucket B3 of tablet **150-3**, the first decref job **180a** reaches the target number **182**, but this occurs before processing the entire B3 column.

(28) In this example, the first decref job **180a** performs a destage operation **166** based on buckets B3 in tablets **150-3** to **150-M** only. The first decref job **180a** thus processes the target number **182** of decrefs and stops there.

(29) Sometime later, when resources become available, the scheduler **170** starts a second decref job **180b**. Job **180b** picks up where job **180a** left off, at bucket B3 of tablet **150-2**. Job **180b** proceeds to the end of the B3 column and performs another destage operation **166**. But having not yet reached the target number **182**, job **180b** proceeds to the B4 column, in this case reaching the target number **182** at the end of the B4 column, whereupon another destage operation **166** is performed.

(30) In the example of FIG. 2, aggregation is slightly reduced, given that two different read-modify-write operations **164** are performed when processing the B3 column. However, the number of columns processed before reaching the target number **182** is typically larger than two or three, such that the impact on aggregation (and thus flash wear) tends to be small.

(31) FIG. 3 shows another way of proceeding when the decref job **180** reaches the target number **182** of decrefs before reaching the end of a column **154**. The FIG.-3 example avoids the decrease in aggregation found in the FIG.-2 example, but at the cost of slightly less consistency in the number of decrefs processed per job.

(32) Here, the decref job **180** starts at the top of the B1 column and proceeds to the B2 column and then to the B3 column, performing a destage operation **166** at the end of each column. As before, the target number **182** of decrefs is reached after processing the B3 bucket in tablet **150-3**. But rather than stopping there, the decref job **180** continues processing decrefs until it reaches the end of the B3 column. The decref job **180** has thus exceeded the target number **182**. In an example, the decref job **180** accounts for this overage by providing feedback **310** to the scheduler **170**, informing the scheduler **170** of the excess decrefs that were processed. The scheduler **170** may then compensate by reducing the number of decrefs to be processed by a future decref job, e.g., by the number of excess decrefs.

(33) FIG. 4 shows yet another example. Here, the decref job **180** may be performed by multiple destagers **410** running in parallel. For example, the decref job **180** may assign destagers **410-1** through **410-3** to respective columns B1 to B3. In an example, the decref job **180** staggers operation of the destagers **410-1** through **410-3** such that the second destager **410-2** starts after the first destager **410-1** has finished counting the decrefs in the B1 column. Likewise, the third destager **410-3** starts after the second destager **410-2** has finished counting the decrefs in the B2 column. This approach avoids a situation in which the target number **182** is reached with multiple columns being partially counted. A variety of parallel processing arrangements are contemplated, and the one described above is just an example.

(34) In an example, the tablets **150** each store certain tracking data on a per-tablet basis, such as average amortization (AA) **420** (**420-1** through **420-M**) and total counts **430** (**430-1** through **430-M**). Although first shown in FIG. 4, the same tracking data may be provided in any of the above-described examples.

(35) In an example, average amortization **420** reflects an average number of decrefs aggregated per virtual block (LI) within a tablet. For example, an average amortization **420** of two for a tablet means that, on average, two decrefs have been recorded for each virtual block represented in the tablet. As will be described, average amortization **420** can be used in determining an optimal size of the aggregation domain **160**. For example, a larger aggregation domain **160** may be provided when average amortization **420** is low, as more tablets are needed to make up for the low average. Likewise, a smaller aggregation domain **160** may be provided when average amortization **420** is high, as fewer tablets are needed. In an example, average amortization **420** is computed based on decrefs written to the in-memory buckets **140** and is propagated to the tablets **150** when they are created.

(36) In an example, counts **430** reflect the total number of decrefs counted per bucket within a tablet **150**. Counts **430** can thus assist the decref job **180** in counting decrefs as it proceeds through the buckets **152**. Counts **430** may also be accumulated upon ingest of decrefs into the in-memory buckets **140**. One should appreciate that counts **430** may be provided as an optimization but are not required, as the decref job **180** may simply count decrefs as it proceeds.

(37) FIG. 5 shows example factors that contribute to a determination of N-Optimal **510**, which represents an optimal number of tablets **150** in the aggregation domain **160**. Because each decref job **180** is arranged to process the target number **182** of decrefs, the number of tablets **150** in the aggregation domain **160** is allowed to vary without being constrained by the operation of decref jobs.

(38) As shown, one factor that contributes to N-Optimal **510** is the average amortization **520** of one or more previous batches of tablets. Such average amortization may be based on the per-tablet average amortization **420** described above. The previous batches are tablets that were processed in previous amortization domains **160**. In some examples, time-series analysis may be used to predict N-optimal based on a series of previous amortization domains.

(39) Another factor that contributes to N-Optimal **510** is the average amortization **520** within a current batch of tablets. For example, the average amortization **420** of tablets may be monitored as new tablets are created and added to a current aggregation domain. Once a desired level of aggregation is reached, the current aggregation domain may be closed off and decref processing on that aggregation domain may begin.

(40) Yet another factor that contributes to N-Optimal **510** is the amount of free space **530** in the decref tier **196**. For example, if free space in the decref tier **196** is running low, the storage system may benefit from fewer tablets in the aggregation domain and from processing decrefs from those tablets more quickly, as doing so will avoid an out-of-space condition in the decref tier.

(41) An additional factor that contributes to N-Optimal **510** is the amount of free space **540** in the storage system **116**. A central reason for processing decrefs is to identify virtual blocks whose reference counts **194b** have fallen to zero, as the physical blocks pointed to by such virtual blocks can be reclaimed, creating additional free space in the storage system. Thus, free space running low in the storage system can mean that the number of tablets in the aggregation domain **160** should be kept small, so that decref processing can proceed promptly, allowing more free space to be created.

(42) The storage system preferably balances the factors **520**, **530**, **540**, and **550** to arrive at an optimal number (N-Optimal) of tablets in the aggregation domain **160**. Such balancing may involve combinatorial logic or machine learning, for example.

(43) FIG. 6 shows example factors that contribute to early destaging **650** from an aggregation domain **160** before a desired number of tablets has been created. Such early destaging may also be referred to herein as an “unordered trigger,” and indicates some circumstances that merit or require immediate destaging, i.e., processing of decrefs from existing tablets by decref jobs **180**.

(44) One factor that contributes to early destaging **650** is low free space **620** in the storage system **116**. As described above, decref processing helps to create more free space. Thus, for example, if free space falls below some low-water mark, such as 5%, 10%, or 15% of capacity, early destaging **650** may be triggered to avoid an out-of-space condition. This may occur even though fewer than the desired number of tablets (N-Optimal) has been created.

(45) Another factor that contributes to early destaging **650** is detecting **630** that host I/O requests **112** are idle. For example, if CPU cores that handle I/O requests **112** are idle for some determined interval of time (e.g., 10 seconds, 20 seconds, etc.), the available capacity from those CPU cores can be used instead for running decref jobs **180**. This arrangement reduces processing debt and may be preferred, even though it may mean less-than-optimal decref aggregation.

(46) An additional factor that contributes to early destaging is detecting **640** that processing debt of other background jobs **180a** exceeds some high-water mark. This is particularly the case for background jobs **180a** that depend upon decref processing, such as decref-to-zero activities and storage reclaim.

(47) FIG. 7 shows an example method **700** that may be carried out in connection with the environment **100** and provides an overview of some of the features described above. The method **700** is typically performed, for example, by the software constructs described in connection with FIG. 1, which reside in the memory **130** of a node **120** and are run by the set of processors **124**. The various acts of method **700** may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in orders different from that illustrated, which may include performing some acts simultaneously.

(48) At **710**, an amortization domain **160** is defined that includes multiple tablets **150** created at respective points in time (e.g., T1, T2, . . . TM). Each of the tablets **150** contains multiple sorted

buckets **152** arranged in a respective sequence of bucket identifiers (e.g., B1 through BN) that is repeated across all of the tablets **150**. The sorted buckets **152** contain decrefs **154** for respective sets of metadata pages **194a**.

(49) At **720**, a decref job **180** is assigned to a specified set of bucket identifiers (e.g., one or more columns **154**), such as the B1 column, the B1 column through the B3 column, or the like.

(50) At **730**, the decref job destages **166** multiple decrefs **154** from the tablets **150** to persistent metadata pages **194a**, beginning with decrefs **154** in buckets **152** having the specified set of bucket identifiers and proceeding until a target number **182** of decrefs has been destaged.

(51) In some examples, the method **700** may be embodied as a computer program product including one or more non-transient, computer-readable storage media **750**, such as a magnetic disk, magnetic tape, compact disk, DVD, optical disk, flash drive, solid state drive, SD (Secure Digital) chip or device, Application Specific Integrated Circuit (ASIC), Field Programmable Gate Array (FPGA), and/or the like. Any number of computer-readable media may be used. The media may be encoded with instructions which, when executed on one or more computers or other processors, perform the process or processes described herein. Such media may be considered articles of manufacture or machines, and may be transportable from one machine to another.

(52) An improved technique has been described for managing reference-count decrements (decrefs) **154**. The technique includes arranging decref jobs **180** to process a constant, or nearly constant, number of decrefs per job. A decref job **180** begins with a specified set of bucket identifiers (one or more columns **154**) common to multiple tablets **150** across an aggregation domain **160** and processes decrefs **154** from the specified bucket identifiers until a target number **182** of decrefs has been reached, continuing to additional bucket identifiers (columns) as needed. Advantageously, the improved technique provides a consistent workload for decref jobs, even though the numbers of decrefs in different buckets can vary substantially. Also, the improved technique can typically level the workload of decref jobs without impacting aggregation, which is allowed to vary independently of the workload per decref job.

(53) Having described certain embodiments, numerous alternative embodiments or variations can be made. For example, although embodiments have been described that involve one or more data storage systems, other embodiments may involve computers, including those not normally regarded as data storage systems. Such computers may include servers, such as those used in data centers and enterprises, as well as general purpose computers, personal computers, and numerous devices, such as smart phones, tablet computers, personal data assistants, and the like.

(54) Further, although features have been shown and described with reference to particular embodiments hereof, such features may be included and hereby are included in any of the disclosed embodiments and their variants. Thus, it is understood that features disclosed in connection with any embodiment are included in any other embodiment.

(55) As used throughout this document, the words “comprising,” “including,” “containing,” and “having” are intended to set forth certain items, steps, elements, or aspects of something in an open-ended fashion. Also, as used herein and unless a specific statement is made to the contrary, the word “set” means one or more of something. This is the case regardless of whether the phrase “set of” is followed by a singular or plural object and regardless of whether it is conjugated with a singular or plural verb. Also, a “set of” elements can describe fewer than all elements present. Thus, there may be additional elements of the same kind that are not part of the set. Further, ordinal expressions, such as “first,” “second,” “third,” and so on, may be used as adjectives herein for identification purposes. Unless specifically indicated, these ordinal expressions are not intended to imply any ordering or sequence. Thus, for example, a “second” event may take place before or after a “first event,” or even if no first event ever occurs. In addition, an identification herein of a particular element, feature, or act as being a “first” such element, feature, or act should not be construed as requiring that there must also be a “second” or other such element, feature or act. Rather, the “first” item may be the only one. Also, and unless specifically stated to the contrary,

“based on” is intended to be nonexclusive. Thus, “based on” should be interpreted as meaning “based at least in part on” unless specifically indicated otherwise. Further, although the term “user” as used herein may refer to a human being, the term is also intended to cover non-human entities, such as robots, bots, and other computer-implemented programs and technologies. Although certain embodiments are disclosed herein, it is understood that these are provided by way of example only and should not be construed as limiting.

(56) Those skilled in the art will therefore understand that various changes in form and detail may be made to the embodiments disclosed herein without departing from the scope of the following claims.

Claims

1. A method of managing reference-count decrements (decrefs), comprising: defining an aggregation domain that includes multiple tablets created at respective points in time, each of the tablets containing multiple sorted buckets arranged in a respective sequence of bucket identifiers that is repeated across all of the tablets, the sorted buckets containing decrefs for respective sets of metadata pages; assigning a decref job to a specified set of bucket identifiers; and destaging, by the decref job, multiple decrefs from the tablets to persistent metadata pages, beginning with decrefs in buckets having the specified set of bucket identifiers and proceeding until a target number of decrefs has been destaged.
2. The method of claim 1, wherein destaging multiple decrefs from the tablets includes: merging decrefs from buckets having a particular bucket identifier across at least some of the tablets; and updating one or more metadata pages based on the merged decrefs.
3. The method of claim 2, wherein the decref job is a first decref job, and wherein merging decrefs from the buckets having the particular bucket identifier includes: ending the first decref job after merging fewer than all buckets having the particular bucket identifier, in response to the target number of decrefs being reached; and starting a second decref job configured to merge remaining buckets having the particular bucket identifier that were not merged by the first decref job.
4. The method of claim 1, wherein destaging, by the decref job, multiple decrefs from the tablets includes completely destaging decrefs from all buckets for each bucket identifier processed by the decref job.
5. The method of claim 4, wherein completely destaging all decrefs for each bucket identifier processed by the decref job causes the target number of decrefs to be exceeded, and wherein the method further comprises providing feedback to a system scheduler, the feedback indicating a number of decrefs actually destaged by the decref job and enabling the scheduler to compensate when scheduling future jobs.
6. The method of claim 1, wherein the set of bucket identifiers includes a plurality of bucket identifiers, and wherein destaging, by the decref job, multiple decrefs includes operating multiple destagers in parallel.
7. The method of claim 1, further comprising tracking an average aggregation of decrefs for each tablet and adjusting the number of tablets in the aggregation domain to achieve a desired level of aggregation.
8. The method of claim 1, wherein the aggregation domain includes a number of tablets that is determined based at least in part on an average aggregation of decrefs achieved during one or more previous destaging operations of previously created tablets.
9. The method of claim 8, wherein the number of tablets in the aggregation domain is further based at least in part on an amount of free space in a decref storage tier configured to store tablets awaiting destaging.
10. The method of claim 8, wherein the method is operated in a data storage system, and wherein the number of tablets in the aggregation domain is further based at least in part on an amount of

free space in the data storage system.

11. The method of claim 1, wherein the method is operated in a data storage system, and wherein the method further comprises: establishing a desired number of tablets in the aggregation domain; and before the desired number of tablets has been created, triggering an early destage from the tablets already created, based on a condition of the data storage system.

12. The method of claim 11, wherein the condition of the data storage system includes host I/O requests received by the data storage system being idle.

13. The method of claim 11 wherein the early-destage is triggered based at least in part on an amount of free storage space in the data storage system falling below a low-water mark.

14. The method of claim 11, wherein the data storage system runs a set of other background jobs that have dependencies on the decref job, and wherein the early-destage is triggered based at least in part on an amount of processing debt from the set of other background jobs exceeding a high-water mark.

15. A computerized apparatus, comprising control circuitry that includes a set of processors coupled to memory, the control circuitry constructed and arranged to: define an aggregation domain that includes multiple tablets created at respective points in time, each of the tablets containing multiple sorted buckets arranged in a respective sequence of bucket identifiers that is repeated across all of the tablets, the sorted buckets containing decrefs for respective sets of metadata pages; assign a decref job to a specified set of bucket identifiers; and destage, by the decref job, multiple decrefs from the tablets to persistent metadata pages, beginning with decrefs in buckets having the specified set of bucket identifiers and proceeding until a target number of decrefs has been destaged.

16. A computer program product including a set of non-transitory, computer-readable media having instructions which, when executed by control circuitry of a computerized apparatus, cause the computerized apparatus to perform a method of managing reference-count decrements (decrefs), the method comprising: defining an aggregation domain that includes multiple tablets created at respective points in time, each of the tablets containing multiple sorted buckets arranged in a respective sequence of bucket identifiers that is repeated across all of the tablets, the sorted buckets containing decrefs for respective sets of metadata pages; assigning a decref job to a specified set of bucket identifiers; and destaging, by the decref job, multiple decrefs from the tablets to persistent metadata pages, beginning with decrefs in buckets having the specified set of bucket identifiers and proceeding until a target number of decrefs has been destaged.

17. The computer program product of claim 16, wherein destaging multiple decrefs from the tablets includes: merging decrefs from buckets having a particular bucket identifier across at least some of the tablets; and updating one or more metadata pages based on the merged decrefs.

18. The computer program product of claim 17, wherein the decref job is a first decref job, and wherein merging decrefs from the buckets having the particular bucket identifier includes: ending the first decref job after merging fewer than all buckets having the particular bucket identifier, in response to the target number of decrefs being reached; and starting a second decref job configured to merge remaining buckets having the particular bucket identifier that were not merged by the first decref job.

19. The computer program product of claim 16, wherein the method further comprises tracking an average aggregation of decrefs for each tablet and adjusting the number of tablets in the aggregation domain to achieve a desired level of aggregation.

20. The computer program product of claim 16, wherein the method is operated in a data storage system, and wherein the method further comprises: establishing a desired number of tablets in the aggregation domain; and before the desired number of tablets has been created, triggering an early destage from the tablets already created, based on a condition of the data storage system.
